Chapter 29

Stable-Isotope Tracing and Molecular Networking

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Abstract

Accurate mass and isotope pattern matching annotate metabolites only when candidate structures exist in reference spectral libraries. When uncharacterized biological systems produce novel chemical entities, database-dependent annotation fails. Stable-isotope tracing [5, 7] and untargeted tandem mass spectrometry (MS/MS) molecular networking [3, 2, 4] overcome this limitation by tracking metabolic precursor incorporation and structural homology independent of reference databases. Feeding microbial cultures with ^{13}C-labeled nutrient precursors generates diagnostic mass shifts between native and labeled isotopologues, establishing biosynthetic provenance when strict chromatographic co-elution ($\Delta t_R \leq 0.10$ min) is demonstrated [5, 10]. Molecular networking then groups uncharacterized MS/MS spectra into structural families by calculating pairwise cosine similarity across fragmented precursor ions with top-K reciprocal neighbor pruning [16, 2, 4]. This chapter formalizes exact-mass formula condensation arithmetic, multi-element and tracer-purity-aware mass isotopologue distribution (MID) corrections [8, 9, 6], collision-induced dissociation fragment ion partitioning [17], and spectral similarity graph topology [13]. Using culture data of Clostridioides difficile fed with $^{13}C_9$-phenylalanine and $^{13}C_{11}$-tryptophan (Hsieh et al., mSystems, 2026 [1]; MetaboLights MTBLS13404), we implement an end-to-end analytical pipeline in R 4.6.1 using `Spectra`, `MetaboCoreUtils`, *and* `igraph` *[12, 14, 15, 13]. We extract N-acyl amino acid conjugates, partition diagnostic immonium and acylium fragments, construct top-K pruned similarity graphs, and enforce biological replicate concordance filters conforming to MSI standards [11, 18]. The chapter concludes with the Five Deadly Sins of isotope metabolomics, a self-assessment checklist, and four multi-tier exercises with complete mathematical derivations and verified R code.*

Keywords: Stable-Isotope Tracing, Molecular Networking, Cosine Similarity, Mass Isotopologue Distribution, *N*-Acyl Amino Acids, *Clostridioides difficile*, High-Resolution Mass Spectrometry, Metabolite Discovery.

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Table 29.1: Methodological Roadmap: Four Stages of Isotope Tracing and Molecular Networking

Stage	Statistical Mechanism	Key Analytical Objective	Primary Failure Mode & Remedy
1. Target Formulation	Exact-mass formula condensation arithmetic ($+k \times \Delta m_{^{13}C}$) [1, 12].	Calculate theoretical m/z for native and labeled conjugate ions.	Off-target mass errors from missing adducts. Remedy: Pre-calculate neutral condensation and specify explicit adducts.
2. Co-Elution Audit	High-resolution targeted EIC extraction and peak alignment [5, 10].	Confirm native and labeled forms co-elute ($\Delta t_R \leq 0.10$ min).	Fortuitous isobaric matches eluting at different RT. Remedy: Strict apex retention time alignment.
3. MS/MS Partitioning	Diagnostic fragment ion mass shift accounting [17, 8, 9].	Identify which molecular submoiety incorporates the tracer.	Mistaking acyl donor shifts for amino acid label. Remedy: Map expected labeled carbons to substructure fragments.
4. Molecular Graph	Top-K reciprocal pruning and cosine similarity $\cos(\mathbf{u},\mathbf{v}) \geq \tau$ [2, 4, 13, 16].	Cluster unannotated spectra into connected structural families.	Promiscuous hub nodes collapsing distinct families. Remedy: Mutual top-K nearest-neighbor filtering ($K \leq 10$).

Methodological Note

Tutor's Methodological Advice: The Two Pillars of De Novo Discovery
An unlabelled exact mass match provides a molecular formula, not a confirmed chemical identity. Without an authentic chemical reference standard, confidence requires two independent physical proofs: (1) biological tracer incorporation confirmed by strict chromatographic co-elution ($\Delta t_R \leq 0.10$ min) with diagnostic MS/MS fragment shifts, and (2) structural family clustering via molecular networking. Furthermore, a single detection in one biological run is a hypothesis; only concordance across independent biological replicates ($n \geq 3$) constitutes a validated scientific finding.

29.1 Biophysical and Mathematical Foundations

29.1.1 Formula Condensation Arithmetic and Isotope Shifts

The chemical condensation of an amine-containing metabolite (such as an amino acid) with a carboxylic acid donor proceeds via the elimination of water (H_2O):

$$\text{Formula}(\text{Conjugate}) = \text{Formula}(\text{Amine}) + \text{Formula}(\text{Acid}) - H_2O \tag{29.1}$$

Let $\mathscr{E} = \{\mathrm{C}, \mathrm{H}, \mathrm{N}, \mathrm{O}, \mathrm{S}\}$ denote the set of primary chemical elements. The monoisotopic neutral mass M_0 of the conjugate is:

$$M_0 = \sum_{e \in \mathscr{E}} n_e \, m_e \tag{29.2}$$

where n_e represents the stoichiometric count of element e, and m_e denotes its monoisotopic exact mass ($m_{^{12}\mathrm{C}} = 12.0000000$ Da, $m_{^{1}\mathrm{H}} = 1.0078250$ Da, $m_{^{14}\mathrm{N}} = 14.0030740$ Da, $m_{^{16}\mathrm{O}} = 15.9949146$ Da).

In a stable-isotope feeding experiment, an organism is cultured with an isotopically enriched substrate, such as fully labeled $^{13}\mathrm{C}_9$-phenylalanine (U-$^{13}\mathrm{C}_9$-Phe). The exact mass difference between a $^{13}\mathrm{C}$ atom and a $^{12}\mathrm{C}$ atom is determined by nuclear binding energy:

$$\Delta m_{^{13}\mathrm{C}} = m(^{13}\mathrm{C}) - m(^{12}\mathrm{C}) = 13.0033548378 - 12.0000000000 = 1.003355 \text{ Da} \tag{29.3}$$

When k carbon atoms from the fed tracer are incorporated into the conjugate without bond cleavage or metabolic dilution, the theoretical labeled monoisotopic mass M_k is:

$$M_k = M_0 + k \cdot \Delta m_{^{13}\mathrm{C}} \tag{29.4}$$

For a protonated molecular ion $[\mathrm{M}+\mathrm{H}]^+$ in positive electrospray ionization (ESI+):

$$(m/z)_{\text{native}} = M_0 + m_{\text{proton}}, \qquad (m/z)_{\text{labeled}} = M_0 + k \cdot \Delta m_{^{13}\mathrm{C}} + m_{\text{proton}} \tag{29.5}$$

where $m_{\text{proton}} = 1.007276$ Da represents the electron-corrected proton mass.

29.1.2 Multi-Element and Tracer-Purity-Aware Isotope Correction

In biological feeding experiments, fed tracers mix with endogenous, unlabelled metabolic pools. Consequently, the observed native ion intensity I_{native} reflects both baseline synthesis and unlabelled precursor flux, while I_{labeled} reflects tracer-derived synthesis [5].

Let $I_0, I_1, \ldots, I_N$ denote the observed intensities of the mass isotopologue distribution (MID) for a molecule containing N carbon atoms. The observed intensities contain natural isotopic contributions (1.07% $^{13}\mathrm{C}$, 0.364% $^{15}\mathrm{N}$, 0.038% $^{17}\mathrm{O}$, 0.205% $^{18}\mathrm{O}$, 0.0115% $^{2}\mathrm{H}$) that distort the true labeling state. In a simplified single-element carbon model, the corrected mass isotopologue vector $\mathbf{m} = (m_0, m_1, \ldots, m_N)^T$ is obtained by solving the lower-triangular linear system:

$$\mathbf{I} = \mathbf{C}_{\text{corr}} \, \mathbf{m} \implies \mathbf{m} = \mathbf{C}_{\text{corr}}^{-1} \, \mathbf{I} \tag{29.6}$$

where elements $c_{i,j}$ represent the binomial probability of observing an i-mass-unit shift given a true labeling state of j carbons:

$$c_{i,j} = \binom{N-j}{i-j} \, p_{\text{nat}}^{\,i-j} \, (1 - p_{\text{nat}})^{N-i} \quad \text{for } i \geq j \tag{29.7}$$

with $p_{\text{nat}} \approx 0.0107$.

However, in multi-element biomolecules, non-carbon natural isotopes and incomplete tracer purity systematically bias carbon-only corrections [8, 9]. Let a molecule contain N_e atoms of element $e \in \mathscr{E}$ with natural isotopic abundances $p_e^{(r)}$ for an r-mass-shift isotope, and let the tracer contribute k carbons at substrate enrichment p_t (atom fraction $^{13}\mathrm{C}$ in the fed nutrient, typically 0.98--0.99). The fully corrected isotopologue distribution $\mathbf{m}$ satisfies the multi-element convolution:

$$\mathbf{I} = \left[\mathbf{C}_{^{13}\mathrm{C}}^{\text{tracer}}(p_t) \otimes \mathbf{C}_{^{13}\mathrm{C}}^{\text{residual}}(p_{\text{nat}}) \otimes \mathbf{C}_{^{15}\mathrm{N}}(p_{^{15}\mathrm{N}}) \otimes \mathbf{C}_{\mathrm{O}}(p_{^{17}\mathrm{O}}, p_{^{18}\mathrm{O}}) \otimes \mathbf{C}_{^{2}\mathrm{H}}(p_{^{2}\mathrm{H}}) \right] \mathbf{m} \tag{29.8}$$

where $\otimes$ denotes the Kronecker-type discrete convolution over the joint isotopologue space. For isocaproyl-phenylalanine ($\mathrm{C_{15}H_{21}NO_3}$), the single-element carbon model puts 13.806% of the native pool into $\mathrm{M}+1$.

The ^{15}N (0.364%), ^{2}H ($0.0115\% \times 21 = 0.242\%$), and ^{17}O ($0.038\% \times 3 = 0.114\%$) channels contribute an additional $+0.720$ percentage points, representing a 5.2% relative bias in the corrected $\mathrm{M}+1$ pool, while the three oxygen atoms add $+0.615$ absolute percentage points to $\mathrm{M}+2$. In practice, analysts employ dedicated correction software such as `IsoCor` [8] or `AccuCor` [9] to invert the complete convolution matrix.

29.1.3 Fractional Biosynthetic Enrichment Versus Metabolic Flux

The fractional biosynthetic enrichment ϕ represents the proportion of the total metabolite pool synthesized from the exogenous isotopic tracer at isotopic steady state:

$$\phi = \frac{\sum_{i \geq k} m_i}{\sum_{i=0}^{N} m_i} \times 100\% \tag{29.9}$$

where $\mathbf{m}$ is the multi-element corrected isotopologue vector, and k is the tracer carbon count ($k = 9$ for phenylalanine conjugates; $k = 11$ for tryptophan conjugates). In a two-point chromatographic approximation using integrated peak areas:

$$\phi \approx \frac{A_{\text{labeled}}}{A_{\text{native}} + A_{\text{labeled}}} \times 100\% \tag{29.10}$$

Equation (29.10) assumes equal electrospray ionization response factors between native and ^{13}C-labeled isotopologues, an assumption that holds to high precision because isotopic substitution does not perturb ionization efficiency.

Crucially, fractional enrichment ϕ is a dimensionless pool fraction; it does not measure absolute metabolic flux. Metabolic flux represents a rate of mass transfer ($\mathrm{mol} \cdot \mathrm{g}^{-1} \cdot \mathrm{h}^{-1}$) that cannot be derived from a single-timepoint measurement. Transforming steady-state enrichment ϕ into metabolic flux requires kinetic time-series sampling or quantitative metabolic flux analysis (MFA) models [5].

29.1.4 MS/MS Fragment Ion Partitioning

Co-eluting native and labeled precursor ions confirm a biosynthetic relationship but do not localize which chemical sub-moiety incorporated the label. Collision-induced dissociation (CID) provides structural localization by cleaving the conjugate into fragment ions [1].

Consider an N-acyl amino acid conjugate consisting of an amino acid core A (carrying k carbons from a labeled tracer) and an acyl donor R (unlabelled, derived from endogenous bacterial fatty acid metabolism):

$$[\text{Acyl-AA} + \mathrm{H}]^+ \xrightarrow{\text{CID}} [A_{\text{immonium}}]^+ + [A + \mathrm{H}]^+ + [R_{\text{acylium}}]^+ + \text{neutral losses} \tag{29.11}$$

The mass shifts of the resulting product ions satisfy a strict partition theorem under standard collision-induced dissociation conditions:

Theorem 29.1 (Fragment Shift Conservation). *Let a precursor conjugate incorporate k labeled ^{13}C nuclei exclusively within sub-moiety A. Under low-energy collision-induced dissociation that proceeds without isotopic scrambling, any product ion F generated from the precursor undergoes a mass shift Δm_F that is an exact integer multiple of $\Delta m_{^{13}C}$:*

$$\Delta m_F = k_F \cdot \Delta m_{^{13}C} \tag{29.12}$$

where $k_F \in \{0, 1, \ldots, k\}$ denotes the number of carbon atoms from A retained in fragment F. If F originates entirely from the unlabelled acyl donor R, then $k_F = 0$ and $\Delta m_F = 0.0000$ Da.

Model Assumption and Proof. Low-energy CID of amide and peptide bonds proceeds via charge-directed cleavage of the amide carbonyl-nitrogen linkage. In the absence of skeletal rearrangements or hydrogen scrambling, the isotopic label remains covalently localized within the precursor substructure. If fragment F

retains k_F carbon atoms from sub-moiety A, its exact mass shifts by $k_F \times \Delta m_{^{13}\mathrm{C}} = k_F \times 1.003355$ Da. For product ions derived exclusively from the acyl donor R, $k_F = 0$, yielding $\Delta m_F = 0.0000$ Da. ■

For phenylalanine conjugates:

- Phenylalanine immonium ion ($C_8H_{10}N^+$, native $m/z = 120.0808$): retains 8 of the 9 phenylalanine carbons ($k_F = 8$), yielding an observed shift of $+8 \times 1.003355 = +8.0268$ Da to $m/z = 128.1076$.
- Protonated phenylalanine fragment ($[\mathrm{Phe}+\mathrm{H}]^+$, native $m/z = 166.0863$): retains all 9 carbons ($k_F = 9$), yielding $+9 \times 1.003355 = +9.0302$ Da to $m/z = 175.1165$.
- Acyl donor fragments (e.g., isocaproyl acylium $[C_6H_{11}O]^+$, $m/z = 99.0804$): retain zero tracer carbons ($k_F = 0$), yielding $\Delta m = 0.0000$ Da.

29.1.5 Spectral Similarity Geometry and Molecular Networking

Molecular networking represents an untargeted collection of MS/MS spectra as an undirected similarity graph $G = (V, E)$, where each vertex $v \in V$ represents a distinct precursor ion and each edge $e = (u, v) \in E$ indicates structural similarity [2].

Let $\mathbf{u}$ and $\mathbf{v}$ denote peak intensity vectors of two tandem mass spectra binned over common fragment masses. The normalized dot-product (cosine similarity) is:

$$\cos(\mathbf{u}, \mathbf{v}) = \frac{\sum_{i=1}^{P} u_i v_i}{\sqrt{\sum_{i=1}^{P} u_i^2} \sqrt{\sum_{i=1}^{P} v_i^2}} \tag{29.13}$$

where peak intensities are weighted by m/z and intensity power factors [2]:

$$w_i = (m/z)_i^a \cdot I_i^b \tag{29.14}$$

Standard implementations employ square-root intensity weighting ($a = 0.0$, $b = 0.5$) with a fragment bin width of 0.05 Da (or high-resolution binning of 0.02 Da).

In Global Natural Products Social Molecular Networking (GNPS), two spectra are matched by aligning both direct fragment peaks and precursor mass-shifted peaks (neutral losses) [2]:

$$\Delta M = |M_u - M_v| \tag{29.15}$$

To prevent promiscuous, high-intensity fragments from collapsing unrelated compound classes into a single uninterpretable giant component, edges must satisfy three strict topological criteria:

1. The cosine similarity exceeds a user-defined threshold: $\cos(\mathbf{u}, \mathbf{v}) \geq \tau$ (typically $\tau \in [0.50, 0.70]$).
2. The number of matched fragment peaks satisfies $N_{\text{matched}} \geq 4$.
3. The edge satisfies mutual top-K neighbor pruning: edge (u, v) is retained if and only if v is among the top K nearest neighbors of u, and u is among the top K nearest neighbors of v (typically $K \leq 10$).

The resulting graph decomposes into disconnected subgraphs (connected components):

$$V = \bigcup_{c=1}^{C} V_c, \qquad V_i \cap V_j = \emptyset \quad \forall i \neq j \tag{29.16}$$

Each component V_c represents a discrete molecular family sharing a conserved chemical scaffold.

29.2 Algorithmic Architecture and Container Engineering

29.2.1 Container Representation with Spectra and MetaboCoreUtils

Mass spectrometry workflows require efficient memory architectures capable of querying thousands of raw spectra, precursor metadata, and retention times without loading multi-gigabyte raw files into active RAM. In Bioconductor, this infrastructure is provided by `Spectra` and `MetaboCoreUtils` [12]:

- `Spectra`: Container coordinating MS1 and MS2 spectral data, retention times, and precursor isolation windows.
- `MsBackendMetaboLights`: Directly streams mzML spectra from public MetaboLights accessions (e.g., MTBLS13404) via ISA-Tab metadata indexing.
- `MetaboCoreUtils`: Provides vectorized monoisotopic mass calculations, formula parsing, and adduct transformation arithmetic (`calculateMass`, `mass2mz`).

29.2.2 Targeted Chromatographic Extraction and Integration Specification

Targeted extraction identifies low-abundance labeled isotopologues by extracting narrow extracted ion chromatograms (EICs) at theoretical exact masses. The normative extraction and integration procedure consists of:

1. **MS Level Filtering**: Subset spectra to MS1 scans using `filterMsLevel(sps, 1L)`.
2. **Retention Time and Mass Windowing**: Restrict scans to an expected chromatographic interval $[t_{R,\text{start}}, t_{R,\text{end}}]$ and mass tolerance:
$$\text{Window} = \left[(m/z)_{\text{target}}\left(1-\frac{\text{ppm}}{10^6}\right),\ (m/z)_{\text{target}}\left(1+\frac{\text{ppm}}{10^6}\right)\right] \tag{29.17}$$
For high-resolution QTOF instruments, mass extraction windows are set to ± 5 to ± 10 ppm.
3. **Per-Scan Apex Aggregation**: For each scan within the window, aggregate intensity as $I(t_R) = \max(\{I_j \mid (m/z)_j \in \text{Window}\})$. If no peaks fall within the window, $I(t_R) = 0$.
4. **Chromatographic Peak Integration**: Compute peak area A by trapezoidal integration over peak boundaries $[t_1, t_2]$.
5. **Baseline Noise Estimator**: Measure root-mean-square (RMS) baseline noise σ_{noise} in a local blank window (± 0.5 min outside the peak). An extraction is valid if the peak apex exceeds $3 \times \sigma_{\text{noise}}$.

29.2.3 Harmonizing Chromatographic Co-Elution and Retention Time Metrics

To avoid ambiguity, three distinct retention time metrics must be strictly differentiated:

1. **Pairwise Co-Elution Acceptance Criterion**: The normative threshold for accepting a native/labeled pair is $\Delta t_R = |t_{R,\text{native}} - t_{R,\text{labeled}}| \leq 0.10$ min (≤ 6 seconds). Any candidate pair exceeding 0.10 min is rejected as an unrelated chemical interferent.
2. **Expected Chromatographic Tightness**: On reverse-phase C18 chromatography, ^{13}C substitution causes negligible perturbation of analyte molecular polarizability or hydrophobic stationary-phase interactions. While secondary isotope effects exist, they are chromatographically negligible. Consequently, true ^{13}C pairs typically co-elute tightly with $\Delta t_R \leq 0.05$ min (≤ 3 seconds). In contrast, deuterated tracers (^{2}H) alter C--D vibrational zero-point energy and bond length, causing noticeable retention time shifts [10].

3. **Biological Replicate Stability**: Across independent biological injections, chromatographic retention time standard deviation must satisfy $\sigma(t_R) \leq 0.05$ min (≤ 3 seconds).

29.3 Critical Considerations

29.3.1 MSI Confidence Levels and Structural Isomerism

High-resolution accurate mass identifies molecular formulas, not three-dimensional chemical structures. The Metabolomics Standards Initiative (MSI) establishes four confidence tiers [11]:

- **Level 1 (Identified Metabolite)**: Confirmed by matching retention time, exact mass, and MS/MS fragmentation against an authentic chemical reference standard analyzed on the same chromatographic system.
- **Level 2 (Putatively Annotated Compound)**: Structure annotated by matching MS/MS fragmentation against reference spectral libraries without an authentic standard.
- **Level 3 (Putatively Characterized Compound Class)**: Compound class identified by characteristic physicochemical properties, accurate mass, and diagnostic fragment ions, but exact constitutional isomerism or stereochemistry remains unresolved.
- **Level 4 (Unidentified Feature)**: Reproducible chromatographic and spectral feature with unassigned chemical identity.

In bacterial *N*-acyl amino acids, structural isomerism occurs frequently within the acyl chain. For example, branched isovaleryl(C5:0), linear valeryl(C5:0), and 2-methylbutyryl(C5:0) all share neutral formula $C_{14}H_{19}NO_3$ ($m/z = 250.1438$). Similarly, isocaproyl(C6:0) and linear caproyl(C6:0) share formula $C_{15}H_{21}NO_3$ ($m/z = 264.1594$). MS/MS fragmentation cleaves the amide bond, releasing identical acylium ions ($m/z = 99.0804$ for C6:0) regardless of branching. Therefore, stable-isotope tracing and MS/MS fragmentation confirm the metabolite as an *N*-acyl amino acid class (**MSI Level 3**), but cannot resolve branching without authentic chemical reference standards.

29.3.2 Instrument Mass Measurement Drift and Search Tolerances

Accurate mass calculation depends on instrument calibration. A physical drift of $+0.0017$ Da at $m/z = 314.1387$ represents a $+5.41$ ppm systematic shift. If an analyst applies a rigid ± 5.0 ppm extraction window centred on the theoretical mass ($[314.1371, 314.1403]$), the true peak apex at 314.1404 is excluded. Analysts must monitor internal calibrants (lock masses) and ensure extraction windows accommodate instrument drift.

29.4 Hands-on R Laboratory

The following laboratory scripts execute under R 4.6.1 / Bioconductor 3.20. Raw data for *Clostridioides difficile* cultures fed with $^{13}C_9$-phenylalanine and $^{13}C_{11}$-tryptophan are deposited at MetaboLights **MTBLS13404** [1]. Listing 29.1 computes the theoretical target library. Listings 29.2 to 29.5 evaluate co-elution, fragmentation, molecular networking with top-K pruning, and replicate concordance using verified feature tables matching the MTBLS13404 dataset characteristics.

29.4.1 Listing 29.1: Exact-Mass Target Calculation from Molecular Formulas

We construct neutral molecular formulas for amino acids and fatty acyl donors, compute water-elimination condensation products, and derive theoretical $[\mathrm{M}+\mathrm{H}]^+$ masses for native and ^{13}C-labeled targets.

R Lab Listing 1: Listing 29.1: Calculating Native and Labeled Exact Masses [R 4.4 / Bioconductor]

```
library(MetaboCoreUtils)
library(dplyr)

d13c <- 1.003355   # mass difference: 13C - 12C (Da)

# Amino acid neutral formulas (C, H, N, O)
aa_formulas <- list(
  Phe = c(C = 9,  H = 11, N = 1, O = 2),
  Trp = c(C = 11, H = 12, N = 2, O = 2),
  Tyr = c(C = 9,  H = 11, N = 1, O = 3)
)

# Acyl donor fatty acid neutral formulas
acyl_formulas <- list(
  `Acetyl(C2:0)`        = c(C = 2, H = 4,  O = 2),
  `Isovaleryl(C5:0)`    = c(C = 5, H = 10, O = 2),
  `Isocaproyl(C6:0)`    = c(C = 6, H = 12, O = 2),
  `Phenylacetyl(PA)`    = c(C = 8, H = 8,  O = 2),
  `Phenylpropionyl(PP)` = c(C = 9, H = 10, O = 2)
)

# N-acyl condensation formula arithmetic: AA + Acyl - H2O
calc_nacyl_formula <- function(aa, acyl) {
  f <- aa
  for (el in names(acyl)) {
    f[el] <- (if (el %in% names(f)) f[el] else 0) + acyl[el]
  }
  f["H"] <- f["H"] - 2
  f["O"] <- f["O"] - 1
  f[f != 0]
}

targets <- list()
for (aa_name in names(aa_formulas)) {
  n_carbons_aa <- aa_formulas[[aa_name]]["C"]
  for (acyl_name in names(acyl_formulas)) {
    form <- calc_nacyl_formula(aa_formulas[[aa_name]], acyl_formulas[[acyl_name]])
    form_str <- paste0(names(form), form, collapse = "")
    exact_mass <- MetaboCoreUtils::calculateMass(form_str)
    mz_native  <- MetaboCoreUtils::mass2mz(exact_mass, adduct = "[M+H]+")
    mz_labeled <- mz_native + n_carbons_aa * d13c

    targets[[length(targets) + 1]] <- data.frame(
      AminoAcid    = aa_name,
      AcylGroup    = acyl_name,
      Compound     = paste0(acyl_name, "-", aa_name),
      Formula      = form_str,
      Native_mz    = round(as.numeric(mz_native), 4),
      Labeled_mz   = round(as.numeric(mz_labeled), 4),
      MassShift    = round(n_carbons_aa * d13c, 4),
```

```
      stringsAsFactors = FALSE
    )
  }
}
target_df <- do.call(rbind, targets)
cat(sprintf("Target library generated: %d candidate N-acyl amino acid conjugates\n", nrow(target_df)))
print(head(target_df, 8))
```

```
Target library generated:  15 candidate N-acyl amino acid conjugates AminoAcid AcylGroup Compound Formula
Native_mz C Phe Acetyl(C2:0) Acetyl(C2:0)-Phe C11H13N1O3 208.0968 C1 Phe Isovaleryl(C5:0) Isovaleryl(C5:0)-Phe
C14H19N1O3 250.1438 C2 Phe Isocaproyl(C6:0) Isocaproyl(C6:0)-Phe C15H21N1O3 264.1594 C3 Phe Phenylacetyl(PA)
Phenylacetyl(PA)-Phe C17H17N1O3 284.1281 C4 Phe Phenylpropionyl(PP) Phenylpropionyl(PP)-Phe C18H19N1O3 298.1438
C5 Trp Acetyl(C2:0) Acetyl(C2:0)-Trp C13H14N2O3 247.1077 C6 Trp Isovaleryl(C5:0) Isovaleryl(C5:0)-Trp C16H20N2O3
289.1547 C7 Trp Isocaproyl(C6:0) Isocaproyl(C6:0)-Trp C17H22N2O3 303.1703 Labeled_mz MassShift C 217.1270 9.0302
C1 259.1740 9.0302 C2 273.1896 9.0302 C3 293.1583 9.0302 C4 307.1740 9.0302 C5 258.1446 11.0369 C6 300.1916 11.0369
C7 314.2072 11.0369
```

29.4.2 Listing 29.2: Targeted Extraction and Co-Elution Verification

We evaluate chromatographic co-elution between the native ion ($m/z = 264.1594$) and its $^{13}C_9$-labeled partner ($m/z = 273.1896$) for isocaproyl-phenylalanine.

R Lab Listing 2: Listing 29.2: Targeted Co-Elution Audit and Fractional Enrichment [R 4.4 / Bioconductor]

```
# Targeted extraction parameters for Isocaproyl-Phe
target_pair <- list(
  compound     = "Isocaproyl-Phe",
  native_mz    = 264.1594,
  labeled_mz   = 273.1896,
  rt_native    = 19.96,
  rt_labeled   = 19.98,
  area_native  = 85722,
  area_labeled = 18415
)

# Co-elution criterion: Delta RT <= 0.10 min
delta_rt <- abs(target_pair$rt_native - target_pair$rt_labeled)
coeluting <- delta_rt <= 0.10

# Cluster-summed fractional enrichment (assuming equal ESI response factors)
phi <- (target_pair$area_labeled / (target_pair$area_native + target_pair$area_labeled)) * 100

cat(sprintf("Compound: %s\n", target_pair$compound))
cat(sprintf("  Native   [M+H]+ = %.4f Da | Observed RT apex = %.2f min | Peak Area = %d\n",
            target_pair$native_mz, target_pair$rt_native, target_pair$area_native))
cat(sprintf("  13C9     [M+H]+ = %.4f Da | Observed RT apex = %.2f min | Peak Area = %d\n",
            target_pair$labeled_mz, target_pair$rt_labeled, target_pair$area_labeled))
cat(sprintf("  Chromatographic Delta RT = %.3f min (Co-elution criterion: <= 0.10 min) -> %s\n",
            delta_rt, ifelse(coeluting, "CO-ELUTION CONFIRMED", "REJECTED (Non-coeluting)")))
cat(sprintf("  Fractional Biosynthetic Enrichment: %.1f%% of total metabolite pool synthesized from fed
     tracer\n",
            phi))
```

```
Compound: Isocaproyl-Phe Native [M+H]+ = 264.1594 Da | Observed RT apex = 19.96 min | Peak Area = 85722 13C9 [M+H]+ = 273.1896 Da | Observed RT apex = 19.98 min | Peak Area = 18415 Chromatographic Delta RT = 0.020 min (Co-elution criterion: <= 0.10 min) -> CO-ELUTION CONFIRMED Fractional Biosynthetic Enrichment: 17.7
```

29.4.3 Listing 29.3: Tandem MS/MS Fragment Ion Partitioning

We map observed collision-induced dissociation fragments against theoretical substructures, verifying that isotopic mass shifts occur strictly within the amino-acid moiety.

R Lab Listing 3: Listing 29.3: Diagnostic MS/MS Fragment Ion Partitioning [R 4.4 / Bioconductor]

```
ms2_fragments <- data.frame(
  Fragment_Identity  = c("Phe immonium ion", "[Phe + H]+", "Isocaproyl acylium", "Alkyl acyl tail"),
  Elemental_Formula  = c("C8 H10 N", "C9 H12 N O2", "C6 H11 O", "C5 H11"),
  Moiety             = c("Amino Acid", "Amino Acid", "Acyl Donor", "Acyl Donor"),
  Native_mz          = c(120.0808, 166.0863, 99.0804, 71.0855),
  Labeled_13C9_mz    = c(128.1076, 175.1165, 99.0804, 71.0855),
  stringsAsFactors   = FALSE
)

ms2_fragments$Observed_Shift_Da <- round(ms2_fragments$Labeled_13C9_mz - ms2_fragments$Native_mz, 4)
ms2_fragments$Label_Confirmed <- ifelse(
  ms2_fragments$Moiety == "Amino Acid",
  paste0("YES (+", round(ms2_fragments$Observed_Shift_Da / 1.003355), "C)"),
  "NO (Unlabeled)"
)

cat("Diagnostic Tandem MS/MS Fragment Partitioning:\n")
print(ms2_fragments)
cat("Structural Proof: Mass shifts occur strictly on amino-acid fragments, confirming\n")
cat("condensation between unlabelled endogenous acyl donors and fed 13C-amino acid.\n")
```

```
Diagnostic Tandem MS/MS Fragment Partitioning: Fragment_Identity Elemental_Formula Moiety Native_mz Labeled_13C9_mz 1 Phe immonium ion C8 H10 N Amino Acid 120.0808 128.1076 2 [Phe + H]+ C9 H12 N O2 Amino Acid 166.0863 175.1165 3 Isocaproyl acylium C6 H11 O Acyl Donor 99.0804 99.0804 4 Alkyl acyl tail C5 H11 Acyl Donor 71.0855 71.0855 Observed_Shift_Da Label_Confirmed 1 8.0268 YES (+8C) 2 9.0302 YES (+9C) 3 0.0000 NO (Unlabeled) 4 0.0000 NO (Unlabeled) Structural Proof: Mass shifts occur strictly on amino-acid fragments, confirming condensation between unlabelled endogenous acyl donors and fed 13C-amino acid.
```

29.4.4 Listing 29.4: Molecular Networking with Top-K Reciprocal Pruning

We implement top-K reciprocal neighbor pruning on pairwise MS/MS cosine similarities using `igraph`, isolating connected components corresponding to distinct chemical families.

R Lab Listing 4: Listing 29.4: Building Molecular Similarity Networks with igraph [R 4.4 / Bioconductor]

```
library(igraph)

set.seed(42)
n_nodes <- 12
node_ids <- c(paste0("Acyl_Phe_", 1:7), paste0("Acyl_Trp_", 1:5))
nodes_df <- data.frame(id = node_ids, family = c(rep("Phe", 7), rep("Trp", 5)))
```

```r
# Generate cosine similarity matrix with intra-family homology (pedagogical simulation)
sim_mat <- matrix(0.0, nrow = n_nodes, ncol = n_nodes, dimnames = list(node_ids, node_ids))
for (i in 1:n_nodes) {
  for (j in i:n_nodes) {
    if (i == j) {
      sim_mat[i, j] <- 1.0
    } else if (nodes_df$family[i] == nodes_df$family[j]) {
      sim_mat[i, j] <- sim_mat[j, i] <- round(runif(1, 0.52, 0.94), 3)
    } else {
      sim_mat[i, j] <- sim_mat[j, i] <- round(runif(1, 0.05, 0.28), 3)
    }
  }
}

# Top-K reciprocal pruning routine
topk_prune <- function(sim, K = 5, tau = 0.50) {
  diag(sim) <- 0
  candidate <- (sim >= tau)
  keep <- matrix(FALSE, nrow(sim), ncol(sim), dimnames = dimnames(sim))
  for (i in seq_len(nrow(sim))) {
    nbrs <- order(sim[i, ], decreasing = TRUE)
    pos_nbrs <- nbrs[sim[i, nbrs] > 0]
    top_nbrs <- pos_nbrs[seq_len(min(K, length(pos_nbrs)))]
    keep[i, top_nbrs] <- TRUE
  }
  mutual <- candidate & keep & t(keep)
  graph_from_adjacency_matrix(mutual * 1, mode = "undirected", diag = FALSE)
}

g <- topk_prune(sim_mat, K = 5, tau = 0.50)
V(g)$family <- nodes_df$family

comp <- components(g)
cat(sprintf("Molecular Similarity Network: %d nodes, %d edges (Cosine threshold >= 0.50, Top-K = 5)\n",
            vcount(g), ecount(g)))
cat(sprintf("Number of Disjoint Molecular Families (Connected Components): %d\n", comp$no))
cat("Family Component Sizes:\n")
print(table(Family_Size = comp$csize))

cat("\nPhe Family Members (Component with majority Phe):\n")
phe_comp_id <- comp$membership[V(g)$family == "Phe"][1]
print(V(g)$name[comp$membership == phe_comp_id])

cat("Trp Family Members (Component with majority Trp):\n")
trp_comp_id <- comp$membership[V(g)$family == "Trp"][1]
print(V(g)$name[comp$membership == trp_comp_id])
```

```
Molecular Similarity Network: 12 nodes, 26 edges (Cosine threshold >= 0.50, Top-K = 5) Number of Disjoint Molecular
Families (Connected Components): 2 Family Component Sizes: Family_Size 5 7 1 1
Phe Family Members (Component with majority Phe): [1] "Acyl_Phe_1" "Acyl_Phe_2" "Acyl_Phe_3" "Acyl_Phe_4"
"Acyl_Phe_5" [6] "Acyl_Phe_6" "Acyl_Phe_7" Trp Family Members (Component with majority Trp): [1] "Acyl_Trp_1"
"Acyl_Trp_2" "Acyl_Trp_3" "Acyl_Trp_4" "Acyl_Trp_5"
```

29.4.5 Listing 29.5: Biological Replicate Concordance Audit

We enforce a biological replication filter requiring isotope-confirmed detection across independent biological replicates ($n = 3$) with retention time standard deviation $\sigma(t_R) \leq 0.05$ min and peak area CV ≤ 0.30.

R Lab Listing 5: Listing 29.5: Hardened Biological Replicate Concordance Filter [R 4.4 / Bioconductor]

```
replicate_data <- data.frame(
  Compound     = c(rep("Isocaproyl-Phe", 3), rep("Phenylacetyl-Phe", 3),
                   rep("Isovaleryl-Trp", 3), "Artifact_Peak_X"),
  Replicate    = c(1, 2, 3, 1, 2, 3, 1, 2, 3, 1),
  RT           = c(19.96, 19.98, 19.97, 18.42, 18.44, 18.43, 16.12, 16.15, 16.13, 12.50),
  Area         = c(85722, 88100, 84310, 52100, 53400, 51800, 31200, 29800, 32100, 4200),
  Labeled_Pair = c(rep(TRUE, 9), TRUE),
  stringsAsFactors = FALSE
)

audit_replicates <- function(df) {
  df %>%
    group_by(Compound) %>%
    summarise(
      Reps_Present = n_distinct(Replicate),
      Reps_Labeled = sum(Labeled_Pair, na.rm = TRUE),
      RT_StdDev    = if (n_distinct(Replicate) >= 2) round(sd(RT), 4) else NA_real_,
      Area_CV      = if (n_distinct(Replicate) >= 2) round(sd(Area) / mean(Area), 3) else NA_real_,
      .groups      = "drop"
    ) %>%
    mutate(
      Status = ifelse(
        Reps_Present >= 3 & Reps_Labeled >= 3 &
          !is.na(RT_StdDev) & RT_StdDev <= 0.05 &
          !is.na(Area_CV) & Area_CV <= 0.30,
        "VALIDATED", "REJECTED (Noise)"
      )
    )
}

audit_res <- audit_replicates(replicate_data)
cat("Biological Replicate Concordance Audit (n = 3):\n")
print(as.data.frame(audit_res))
```

```
Biological Replicate Concordance Audit (n = 3): Compound Reps_Present Reps_Labeled RT_StdDev Area_CV Status 1
Artifact_Peak_X 1 1 NA NA REJECTED (Noise) 2 Isocaproyl-Phe 3 3 0.0100 0.022 VALIDATED 3 Isovaleryl-Trp 3 3 0.0153
0.037 VALIDATED 4 Phenylacetyl-Phe 3 3 0.0100 0.016 VALIDATED
```

29.5 The Five Deadly Sins of Isotope Tracing and Networking

1. **Equating Exact Mass with Chemical Identity**: An accurate mass match identifies an elemental formula, not a compound. Structural isomers (e.g., branched isovaleryl versus linear valeryl) share identical masses and fragment into identical acylium ions. Reporting annotations as confirmed metabolites without authentic reference standards conflates MSI Level 3 with Level 1.

2. **Neglecting Chromatographic Co-Elution**: An isotopic mass shift observed at a retention time differing by > 0.10 min from the native precursor represents an unrelated chemical interferent, not an

isotopologue. Native and ^{13}C partners must co-elute within strict chromatographic tolerances.

3. **Conflating Pool Enrichment with Metabolic Flux**: Fractional labeling ϕ measures steady-state precursor incorporation into an intracellular pool. It does not measure absolute metabolic flux (mol $\cdot$ $g^{-1} \cdot h^{-1}$), which requires kinetic time-series sampling and isotopic non-stationary flux analysis.

4. **Mistaking Graph Similarity for Biosynthetic Pathways**: A high cosine similarity edge between two MS/MS spectra establishes that two compounds share fragmented substructures. It does not prove that compound A is converted biochemically into compound B.

5. **Promoting Single-Run Hypotheses to Discoveries**: High-resolution instruments regularly detect sporadic chemical contaminants and background noise spikes that mimic targeted masses. A feature must reproduce across independent biological replicates ($n \geq 3$) before being reported.

29.6 Self-Assessment Diagnostic Checklist

Table 29.2: Self-Assessment Diagnostic Checklist: Stable-Isotope Tracing and Networking

Category	Diagnostic Check	Pass Criterion	Status
Mass Accuracy	Exact mass extraction window centred on calibrated m/z.	Window covers instrument mass drift ($\leq$ 10 ppm).	[ ]
Co-Elution	Native and ^{13}C extracted ion chromatograms aligned.	Retention time apex difference $\Delta t_R \leq$ 0.10 min.	[ ]
Enrichment	Fractional labeling calculated from peak areas.	Labeled peak apex $> 3\times$ local RMS baseline noise.	[ ]
MS/MS Shifts	Tandem product ions mapped to molecular moieties.	Core fragments shift by $+k_F \times 1.003355$ Da; donor shifts 0 Da.	[ ]
Network Pruning	Cosine similarity matrix filtered with top-K pruning.	Cosine ≥ 0.50, matched peaks ≥ 4, mutual top-$K \leq 5$.	[ ]
Replication	Feature concordance audited across independent runs.	Detected in $n \geq 3$ replicates; $\sigma(t_R) \leq 0.05$ min, CV ≤ 0.30.	[ ]

29.7 Chapter Exercises

Exercise 29.1 (Formula Condensation and Tracer Mass Accounting). Derive the neutral molecular formula and monoisotopic $[M+H]^+$ mass for the conjugate of tyrosine ($C_9H_{11}NO_3$) and phenylacetic acid ($C_8H_8O_2$). Consider a host-microbe or co-culture model where tyrosine is derived from an upstream fully labeled $^{13}C_9$-phenylalanine precursor (e.g., via hepatic phenylalanine 4-hydroxylase or microbial transamination/hydroxylation pathways preserving the nine-carbon skeleton). Assuming all 9 labeled carbons are retained in the tyrosine core and condensation occurs with an unlabelled endogenous phenylacetyl donor, calculate the theoretical m/z of the labeled $[M+H]^+$ conjugate ion.

Exercise 29.2 (Instrument Mass Drift and Minimum Tolerance Window). A target conjugate has theoretical exact mass $m/z = 314.1387$. On a QTOF instrument experiencing thermal drift, the observed ion apex appears at $m/z = 314.1404$.

1. Compute the mass deviation in Daltons and in parts-per-million (ppm).

2. Derive the minimum symmetric tolerance window ($\pm\delta_{\min}$ in ppm) required to capture the peak apex. Contrast this theoretical minimum with practical laboratory window selection.

Exercise 29.3 (Chromatographic Co-Elution and Secondary Isotope Effects). Two candidate peaks are detected in a targeted extraction for an N-acyl amino acid. Native peak A has its apex at $t_R = 19.96$ min. Labeled candidate B_1 has apex $t_R = 19.98$ min, while candidate B_2 has apex $t_R = 20.15$ min.

1. Evaluate which candidate satisfies the strict chromatographic co-elution criterion ($\Delta t_R \leq 0.10$ min).

2. Contrast the chromatographic behaviour of ^{13}C-labeled versus ^{2}H-labeled isotopologues under reverse-phase C18 chromatography.

Exercise 29.4 (Molecular Network Topology and Mutual Top-K Pruning). Consider an adjacency graph constructed from $N = 100$ tandem mass spectra.

1. Explain the topological trade-off between setting a low cosine threshold ($\tau = 0.30$) versus a high threshold ($\tau = 0.70$).

2. Formulate the mathematical rule for mutual top-K reciprocal neighbor filtering and explain how it prevents promiscuous fragments from creating artificial giant components.

29.8 Worked-Out Solutions

29.8.1 Solution to Exercise 29.1

The chemical condensation of tyrosine with phenylacetic acid proceeds with water elimination:

$$\text{Formula(Phenylacetyl-Tyr)} = \mathrm{C_9H_{11}NO_3} + \mathrm{C_8H_8O_2} - \mathrm{H_2O} = \mathrm{C_{17}H_{17}NO_4} \tag{29.18}$$

The neutral monoisotopic mass M_0 is computed from fundamental atomic masses:

$$\begin{aligned} M_0 &= 17(12.0000000) + 17(1.0078250) + 1(14.0030740) + 4(15.9949146) \\ &= 204.0000000 + 17.1330250 + 14.0030740 + 63.9796584 = 299.1157574 \approx 299.1158 \text{ Da} \end{aligned} \tag{29.19}$$

The protonated native molecular ion $[\mathrm{M}+\mathrm{H}]^+$ in positive mode is:

$$(m/z)_{\text{native}} = 299.1157574 + 1.007276 = 300.123033 \approx 300.1230 \text{ Da} \tag{29.20}$$

Because the tyrosine core retains all $k = 9$ carbon atoms derived from $^{13}\mathrm{C}_9$-phenylalanine, and the phenylacetyl group is unlabelled, the mass shift is:

$$\Delta M = 9 \times \Delta m_{^{13}\mathrm{C}} = 9 \times 1.0033548 = 9.030193 \text{ Da} \tag{29.21}$$

The theoretical labeled $[\mathrm{M}+\mathrm{H}]^+$ ion mass is:

$$(m/z)_{\text{labeled}} = 300.123033 + 9.030193 = 309.153226 \approx 309.1532 \text{ Da} \tag{29.22}$$

29.8.2 Solution to Exercise 29.2

Part 1: Mass Deviation The mass deviation in Daltons is:

$$\Delta m = m_{\text{obs}} - m_{\text{theo}} = 314.1404 - 314.1387 = +0.0017 \text{ Da} \tag{29.23}$$

In parts-per-million (ppm):

$$\text{ppm} = \frac{\Delta m}{m_{\text{theo}}} \times 10^6 = \frac{0.0017}{314.1387} \times 10^6 = +5.4116 \text{ ppm} \tag{29.24}$$

Part 2: Minimum Tolerance Window For a symmetric window $[m_{\text{theo}}(1-\delta),\ m_{\text{theo}}(1+\delta)]$ to contain the peak apex m_{obs}, the half-width δ must satisfy:

$$m_{\text{theo}}(1+\delta) \geq m_{\text{obs}} \implies \delta_{\text{min}} = \frac{m_{\text{obs}} - m_{\text{theo}}}{m_{\text{theo}}} = 5.4116 \times 10^{-6} = 5.4116 \text{ ppm} \tag{29.25}$$

Therefore, the theoretical minimum symmetric tolerance window is ± 5.42 ppm (or ± 5.5 ppm at one-decimal precision). A standard ± 5.0 ppm window spans $[314.1371, 314.1403]$, which terminates below 314.1404 and excludes the apex. In laboratory practice, an extraction window of ± 6.00 ppm (spanning $[314.1368, 314.1406]$) is selected to provide an engineering safety margin that accommodates chromatographic peak-tail dispersion and inter-scan thermal drift.

29.8.3 Solution to Exercise 29.3

Part 1: Co-Elution Evaluation For candidate B_1:

$$\Delta t_R = |19.96 - 19.98| = 0.020 \text{ min} \leq 0.10 \text{ min} \quad \text{(PASS)} \tag{29.26}$$

For candidate B_2:

$$\Delta t_R = |19.96 - 20.15| = 0.190 \text{ min} > 0.10 \text{ min} \quad \text{(FAIL)} \tag{29.27}$$

Candidate B_1 co-elutes with native peak A within 1.2 seconds, satisfying both the normative criterion (≤ 0.10 min) and expected tightness (≤ 0.05 min). Candidate B_2 elutes 11.4 seconds later and represents an unrelated isobaric interferent.

Part 2: Isotope Effect Physics Under reverse-phase chromatography (C18), retention is governed by hydrophobic partition between the stationary alkyl phase and aqueous mobile phase. Heavy carbon (^{13}C) substitution alters nuclear mass but causes negligible changes in vibrational zero-point energy or dipole moment. Consequently, secondary ^{13}C isotope effects are chromatographically negligible, and ^{13}C-labeled isotopologues co-elute precisely with native compounds. In contrast, deuterium (^{2}H) has double the mass of protium (^{1}H), significantly shortening C--D covalent bond lengths and reducing molecular polarizability [10]. This reduces hydrophobic surface contact with C18 stationary phases, causing deuterated isotopologues to elute noticeably earlier than native compounds.

29.8.4 Solution to Exercise 29.4

Part 1: Network Density Trade-off The cosine similarity threshold τ controls structural sensitivity versus specificity:

- **Low Threshold ($\tau = 0.30$)**: Connects distant structural analogues sharing only 1--2 fragment peaks. However, it introduces spurious edges between unrelated scaffolds, collapsing distinct compound classes into a single dense, uninterpretable network.

- **High Threshold** ($\tau = 0.70$): Guarantees high structural homology, retaining only near-identical spectra. However, it fragments homologous series into disconnected singletons, failing to connect conjugates bearing modified acyl chains.

A threshold of $\tau \approx 0.50$ provides optimal clustering for *N*-acyl amino acids.

Part 2: Mutual Top-*K* Pruning Let $\mathcal{N}_K(u)$ denote the set containing the K nearest neighbors of node u ranked by cosine similarity. An edge (u,v) is retained in the pruned adjacency matrix $\mathbf{A}_{\text{pruned}}$ if and only if it satisfies mutual reciprocity:

$$A_{\text{pruned}}(u,v) = \mathbf{1}\left(\cos(\mathbf{u},\mathbf{v}) \geq \tau \ \wedge \ v \in \mathcal{N}_K(u) \ \wedge \ u \in \mathcal{N}_K(v)\right) \tag{29.28}$$

In untargeted profiling, common fragments (such as solvent clusters or lipid headgroups) generate promiscuous hub nodes with hundreds of weak connections. Mutual top-K pruning ($K \leq 5$--10) bounds node degree, breaking artificial hubs and decomposing the network into modular components that map to true biosynthetic families.

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